

Problem 1

- (1) $\Theta(n^2)$
- (2) $\Theta(n^3)$
- (3) $\Theta(\log n)$
- (4) $\Theta(2^n)$

Problem 2

a. Best-case running time:

- (1) Line 2 executes 1 time, Line 3 executes 1 time, and Line 4 executes 0 times.
- (2) Input condition: The first two elements in the array are out of order, meaning $A[0] > A[1]$.
- (3) $\Theta(1)$

b. Worst-case running time:

- (1) Line 2 executes $n-1$ times, Line 3 executes 0 times, and Line 4 executes 1 time.
- (2) Input condition: The entire array is sorted in nondecreasing order, so the check at Line 2 is never satisfied.
- (3) $\Theta(n)$

Problem 3

a. Best-case running time:

- $\Theta(1)$
- **Input condition:** The integer x is neither even nor divisible by 3 (e.g., $x=1, 5, 7$).
- **Justification:** The initial conditions $x \bmod 2 == 0$ and $x \bmod 3 == 0$ both evaluate to false. The program skips all loops and directly returns the initial result of 0 at Line 9.

b. Worst-case running time:

- $\Theta(n^2)$

- **Input condition:** The integer x is an odd multiple of 3 (e.g., $x=3, 9, 15$).
- **Justification:** This condition bypasses the first loop but triggers the nested loops in the else if block. The outer loop runs n times and the inner loop runs i times, resulting in $n(n+1)/2$ operations.

Problem 4

a. Trace MYSTERY:

- **Iteration 1:** left = 0, right = 6, $A[\text{left}] = 1$, $A[\text{right}] = 15$, sum = 16
- **Iteration 2:** left = 0, right = 5, $A[\text{left}] = 1$, $A[\text{right}] = 10$, sum = 11
- **Iteration 3:** left = 1, right = 5, $A[\text{left}] = 3$, $A[\text{right}] = 10$, sum = 13
- **Iteration 4:** left = 1, right = 4, $A[\text{left}] = 3$, $A[\text{right}] = 8$, sum = 11
- **Iteration 5:** left = 2, right = 4, $A[\text{left}] = 4$, $A[\text{right}] = 8$, sum = 12
- **Return value:** true

b. Problem Statement:

- **Input:** An array A of n elements sorted in nondecreasing order, and a target value x .
- **Output:** Returns true if there are two distinct elements in the array that sum to exactly x ; otherwise, it returns false.

c. Best-case running time:

- $\Theta(1)$
- **Input condition:** The sum of the first and last elements in the array equals x exactly (i.e., $A[0] + A[n-1] == x$).
- **Justification:** The algorithm checks the sum of the first and last elements during the very first iteration of the while loop. It finds the match immediately and returns true, running in constant time.

d. Worst-case running time:

- $\Theta(n)$

- **Input condition:** No two elements in the array sum to x , or the only two elements that sum to x are located at the exact center of the array.
- **Justification:** The left and right pointers must converge by traversing the entire array one step at a time until they meet. The while loop executes approximately $n/2$ times before the $left < right$ condition evaluates to false.

Problem 5

I did not use any Generative AI for this homework assignment, I used the slides provided on folio as a guide.